FOR A SEPARABLE ALGEBRAIC EXTENSION THE MULTIPLICATION MAP IS A LOCALISATION OF THE SQUARE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted statement and proof refer to results of the Fields chapter by label, one of them a section; they are recorded as navigation stubs with the appropriate kinds, and the arguments using them stay inside the counted span.

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Allowed standard prerequisites (editorial).

- The primitive element theorem for finite separable extensions. Invoked in the counted proof at line 10447. Writes a finite separable extension as generated by one element with a separable minimal polynomial.
- Colimits and exactness of localisation. Used in the second half of the counted proof (lines 10460-10471). Passes the finite case to arbitrary separable algebraic extensions and shows the kernel is killed.

Source reference. Original lemma 030N: sources/stacks-chow/tags.txt, source lines 3954–3954. Navigation pointer for \ref{fields-lemma-primitive-element} used inside the counted span (line 10447); the argument that consumes it is retained in the span.

Source reference. Original section 09GZ: sources/stacks-chow/tags.txt, source lines 11987–11987. Navigation pointer for \ref{fields-section-separable-extensions} used inside the counted span (line 10450); the argument that consumes it is retained in the span.

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let k'/k be a separable algebraic extension. Then there exists a multiplicative subset $S \subset k' \otimes_k k'$ such that the multiplication map $k' \otimes_k k' \to k'$ is identified with $k' \otimes_k k' \to S^{-1}(k' \otimes_k k')$.*

Proof. First assume k'/k is finite separable. Then $k' = k(\alpha)$, see Fields, Lemma 030N. Let $P \in k[x]$ be the minimal polynomial of α over k. Then P is an irreducible, separable, monic polynomial, see Fields, Section 09GZ. Then $k'[x]/(P) \to k' \otimes_k k'$, $\sum \alpha_i x^i \mapsto \alpha_i \otimes \alpha^i$ is an isomorphism. We can factor $P = (x - \alpha)Q$ in $k'[x]$ and since P is separable we see that $Q(\alpha) \neq 0$. Then it is clear that the multiplicative set S' generated by Q in $k'[x]/(P)$ works, i.e., that $k' = (S')^{-1}(k'[x]/(P))$. By transport of structure the image S of S' in $k' \otimes_k k'$ works.

In the general case we write $k' = \bigcup k_i$ as the union of its finite subfield extensions over k. For each i there is a multiplicative subset $S_i \subset k_i \otimes_k k_i$ such that $k_i = S_i^{-1}(k_i \otimes_k k_i)$. Let S be the multiplicative closure of $\bigcup S_i \subset k' \otimes_k k'$. Clearly, the multiplication maps sends every element of S to an invertible element of k'. Using that $k' \otimes_k k'$ is the union of the rings $k_i \otimes_k k_i$ we see that every element in the kernel of the multiplication map is mapped to zero in $S^{-1}(k' \otimes_k k')$. Using exactness of localization the result follows. $\square$